

Sumith Reddi Baddam

[LinkedIn](#) | [GitHub](#) | [Homepage](#) | [ResearchGate](#) | [Mail](#) |

PROFESSIONAL SUMMARY

I am a **Senior Applied Scientist at Amazon Sponsored Products** where I currently lead cross-platform advertising. My work **pioneered Amazon's first integrations with Meta Ads platforms**, establishing novel frameworks for real-time bidding, cost-per-click estimation, and incrementality measurement across external advertising ecosystems **serving 2B+ daily active users** and **published research at Amazon Machine Learning Conference on cross-platform advertising optimization**. I am also a **peer-reviewer at AMLC 2025 conference and Consumer Science Summit 2026**. I have a **patent pending with Amazon Patent Group on my work related to dynamic cost-sharing between advertisers and e-commerce platforms in cross-platform advertising which is built into production and is generating \$1B+ in annual incremental sales for sellers nationwide**. Previously, I worked at Amazon Alexa, where I enhanced speech recognition for Agentic AI based voice shopping experiences. I architected error-aware reformulation systems using transformer architectures that achieved +41% error correction rates, positively impacting 100K+ daily customer interactions. Before joining Amazon, I worked as a Data Scientist at Cisco Systems, developing ensemble deep learning models for software defect prediction and have a research publication at *Advances in Economics and Business* 2018. I received dual Masters degrees: Master of Science in Data Science from Indiana University Bloomington in 2020 and Master of Technology in Information Technology from IIIT Bangalore in 2017.

Publications & Patent

- **Measurement and Optimization of Cross-Platform Advertising across Amazon Sponsored Products and Meta Ads**
Sumith Reddi Baddam, et al. *Amazon Machine Learning Conference (AMLC)*, 2025
- **Customer Success Using Deep Learning** [DOI: 10.13189/aeb.2018.060607](#)
Sumith Reddi Baddam, et al. *Advances in Economics and Business*, 2018
- **System and Methods for co-bidding and dynamic cost-sharing between advertisers and E-commerce platforms in Cross-channel advertising (PATENT)**
USPTO Application (Accepted by Amazon Patent Group, In Progress)
Production system generating \$1B+ in incremental sales annually
- **Open source contribution:** Image2Ingredients – Deep learning application integrated into Snapchat Lens (Collaboration with Snap Inc.) [Paper](#) | [GitHub Repository](#)

WORK EXPERIENCE

Senior Applied Scientist

Amazon Ads

Oct 2023 – Present

Arlington, VA

- Pioneered Amazon's [Sponsored Products](#) cross-platform advertising integration with Meta Ads, establishing novel state of the art frameworks for seller expansion into external advertising ecosystems with 2B+ daily active users, resulting in nationwide sellers generating \$1B+ in incremental sales annually. [Press link 1](#) | [Press link 2](#)
- Invented real-time cost-per-click estimation algorithm solving fundamental conflict between Amazon's real-time advertiser pricing model and external platforms' delayed reporting, incorporating dynamic bidding with Bayesian conversion rate estimation. Designed statistically robust experimentation frameworks methodologies for measuring incrementality. Published research at Amazon Machine Learning Conference 2025 on cross-platform advertising optimization.

- Transformed Amazon’s ad sourcing on Pinterest from keyword-based sourcing to visual sourcing, implementing multi-modal AI systems combining image and contextual features for 10M+ daily PINs.
- Architected end-to-end production systems using AWS SageMaker and distributed computing for embedding generation, similarity computation across 20M+ products sold on Amazon achieving +23% increase in conversion rate and +18% return on advertiser spend.

Data Scientist

April 2021 – Oct 2023

Amazon Alexa

Seattle, WA

- Architected error-aware speech recognition system using Text-to-Text Transfer Transformer architectures, achieving +41.34% error correction rate and reducing word error rate by +17%. Deployed production system impacting 100K+ daily customer interactions with Alexa.
- Developed speaker embedding models using Recurrent Neural Network (RNN) models to generate user’s speaker profiles, enabling personalized speech recognition and synthetic voice generation.
- Led comprehensive benchmarking of Alexa speech recognition models against industry leaders OpenAI Whisper and Suno AI Bark, establishing new standards for competitive analysis and identifying critical performance gaps in Agentic AI based voice shopping speech recognition experiences.

Software Development Engineer

Feb 2021 – April 2021

AWS CloudFormation

Seattle, WA

- Developed predictive ensemble machine learning model to estimate cloud infrastructure setup and resource allocation time for AWS foundational service for implementing Infrastructure as Code (IaC).

Data Scientist

Jul 2017 – Jul 2019

Cisco Systems

Bangalore, India

- Developed ensemble Recurrent Neural Networks and Convolutional Neural Network models achieving 80% accuracy in predicting software defects for Cisco routers and switches.
- Published research in peer-reviewed journal Advances in Economics and Business (2018), contributing to software quality improvement methodologies

Invited Talks & Presentations

- **Peer Reviewer:** Amazon Machine Learning Conference (AMLC) 2025, Consumer Science Summit 2026
- **Measurement and Optimization of Cross-Platform Advertising**
Amazon Machine Learning Conference (AMLC), 2025
- **Customer Success using Deep Learning**
Fifth International Conference on Business Analytics and Intelligence (ICBAI), 2017

EDUCATION

Indiana University Bloomington

Bloomington, IN, USA

Master of Science in Data Science

Aug. 2019 – Dec. 2020

International Institute of Information Technology (IIIT), Bangalore

Bangalore, India

Master of Technology in Information Technology

Aug. 2012 – Jul. 2017

Bachelor of Technology in Information Technology

Aug. 2012 – Jul. 2017

Technical Skills

Machine Learning: Natural Language Processing, Computer Vision, Multi-modal AI, Reinforcement Learning, Bayesian Methods, Statistical Modeling, Causal Inference

Deep Learning / AI Frameworks: PyTorch, Tensorflow, Keras, PyRobot, Langchain, OpenAI

Languages: Python, Java, SQL, C++, C, R, MATLAB, Javascript, HTML, CSS

Developer Tools: Git, Docker, Kubernetes, Shell, AWS, Google Cloud Platform, Selenium, Jenkins, Spinnaker, Streamlit